Gradient Descent Algorithm

A Comprehensive Guide

This document provides a detailed explanation of the gradient descent algorithm, including its mathematics, different types, and practical applications.

Autor: Maru

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Introduction

Gradient descent is a fundamental optimization algorithm used in machine learning and statistics to minimize a function by iteratively moving towards the steepest descent direction. It is widely used for training models, particularly in linear regression and neural networks. This document explores the mathematics behind gradient descent, its various types, and includes examples with formulas and tables.

What is Gradient Descent?

Gradient descent minimizes a cost function by adjusting parameters in the direction of the negative gradient. The gradient points to the direction of the steepest ascent, so moving opposite to it reduces the function value. The algorithm repeats this process until convergence.

Mathematics Behind the Algorithm

Consider a cost function J(θ), where θ represents the parameters. The goal is to find θ that minimizes J(θ). The gradient ∇J(θ) indicates the direction of increase. The update rule is:

$$\theta := \theta - \alpha \nabla J(\theta)$$

Gradient Descent Update Rule

Here, α is the learning rate, controlling the step size. For linear regression, the cost function is the mean squared error:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

Mean Squared Error Cost Function

The gradient with respect to θ_j is:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

Partial Derivative of Cost Function

Types of Gradient Descent

There are several variants of gradient descent, differing in how much data is used to compute the gradient at each step. The main types are:

- **Batch Gradient Descent**: Uses the entire dataset to compute the gradient.
- **Stochastic Gradient Descent (SGD)**: Uses one training example at a time.
- **Mini-Batch Gradient Descent**: Uses a small batch of examples.

Comparison of Types

Comparison of Gradient Descent Types

Type	Advantages	Disadvantages
Batch GD	Stable convergence; accurate gradient	Slow for large datasets; high memory usage
SGD	Fast updates; can escape local minima	Noisy updates; may not converge smoothly
Mini-Batch GD	Balances speed and stability	Requires tuning batch size

Detailed Explanations

Batch Gradient Descent

In batch gradient descent, the gradient is computed using all training examples. The update is performed once per epoch. This leads to stable but potentially slow convergence for large datasets.

Stochastic Gradient Descent

SGD updates parameters after each example, making it faster and suitable for online learning. However, the updates are noisy, leading to oscillations around the minimum.

Mini-Batch Gradient Descent

This hybrid approach uses a subset of data (e.g., 32-256 examples) for each update. It reduces noise compared to SGD while being more efficient than batch GD.

Applications

Gradient descent is used in various machine learning tasks, including linear and logistic regression, neural network training, and support vector machines. It powers algorithms like backpropagation in deep learning.

Conclusion

Gradient descent is a powerful tool for optimization in machine learning. Understanding its types and mathematics allows for better model training. Choosing the right variant depends on the dataset size and computational resources.